

Week 9 – System Requirements Document

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Week: Week 9 – Designing & Prototyping Human-Centred Systems (HCI/HRI)

Chosen Implementation: AI-Assisted Emergency Department Triage Dashboard (HCI)

1) System Vision

The proposed system is an AI-assisted Emergency Department (ED) triage dashboard designed to support nurses during the patient triage process. The system analyses patient information entered during triage and provides a recommended Emergency Severity Index (ESI) level together with a confidence score and the key clinical factors influencing the recommendation.

The dashboard is intended to support clinical decision-making rather than replace it. The triage nurse remains responsible for reviewing the recommendation, accepting or overriding the prediction, and making the final clinical decision. The system is designed to integrate into the existing Mercer General Hospital triage workflow while improving consistency, transparency and patient safety.

2) System Inputs

The system receives the following information during patient assessment.

Input	Source
Patient demographics	Manual entry (future EHR integration)
Presenting complaint	Triage nurse
Blood pressure	Manual entry (future medical device integration)
Heart rate	Manual entry (future medical device integration)
Respiratory rate	Manual entry

Body temperature	Manual entry
Oxygen saturation (SpO ₂)	Manual entry (future medical device integration)
Pain score	Manual entry
Clinical observations	Triage nurse

3) System Processing

After receiving patient information, the system performs the following operations:

- Validates all required input fields.
- Identifies missing or incomplete information.
- Processes the patient data using the trained AI triage model.
- Predicts the recommended Emergency Severity Index (ESI).
- Calculates the confidence level of the prediction.
- Identifies the clinical factors that contributed to the recommendation.

4) System Outputs

The system provides the following outputs to the triage nurse.

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Output	Purpose
Recommended ESI Level	Clinical decision support
Confidence Score	Indicates confidence in prediction
Colour-coded Priority Indicator	Rapid visual identification of urgency
Key Contributing Clinical Factors	Improves explainability and transparency
Nurse Override Option	Maintains human oversight
Audit Log	Records recommendations and clinical decisions

5) Functional Requirements

The system shall:

- Display the predicted ESI classification within **1.5 seconds** of data submission.
- Display the AI confidence score for every prediction.
- Present the key clinical factors influencing the recommendation.
- Allow nurses to accept or override the AI recommendation.
- Require an override reason before the final decision is submitted.
- Record recommendations, overrides and timestamps within an audit log.
- Support manual entry of patient information.
- Display patient information in a single dashboard to minimise navigation during triage.

6) Non-Functional Requirements

The system shall:

- Remain clearly legible under hospital lighting conditions.
- Support efficient use during busy Emergency Department workflows.
- Minimise cognitive workload through a simple and intuitive interface.
- Continue operating during periods of high patient volume.
- Protect patient confidentiality and privacy.
- Provide clear visual indicators for high-priority patients.
- Maintain high system availability during clinical use.
- Allow manual clinical workflows if AI recommendations become unavailable.

7) Integration Requirements

The system shall:

- Support future integration with the Mercer General Hospital Electronic Health Record (EHR).
- Support future integration with medical devices such as blood pressure monitors, pulse oximeters and thermometers.
- Connect securely to hospital authentication systems.
- Store audit logs securely for governance and quality assurance.
- Allow manual data entry whenever EHR connectivity is unavailable.
- Support future reporting and performance monitoring.

8) Human Interaction Requirements

The triage nurse shall:

- Review the AI recommendation.
- Review the confidence score and contributing clinical factors.
- Accept or override the recommendation.
- Provide a reason when overriding the AI recommendation.
- Make the final clinical triage decision.

The AI system shall:

- Provide recommendations only.
- Never make the final triage decision.
- Clearly indicate that all recommendations are advisory.

9) Safety Requirements

The system shall:

- Never automatically assign a patient's final triage category.
- Clearly display that recommendations are intended to support, not replace, clinical judgement.
- Record all override decisions within the audit log.
- Flag incomplete or missing patient information before generating recommendations.
- Support manual clinical decision-making at all times.
- Prioritise patient safety by ensuring that human oversight remains mandatory.

10) Summary

The proposed AI-assisted Emergency Department triage dashboard has been designed to integrate into the existing Mercer General Hospital workflow while maintaining human-centred clinical decision-making. The interface provides timely AI recommendations together with transparent supporting information, while ensuring that nurses retain responsibility for every final triage decision. The system also incorporates audit logging, secure data handling and future integration with hospital information systems to support safe and accountable deployment.